

Program : Diploma in Wood and Paper Technology	
Course Code : 6121A	Course Title: Interior Design and Furniture Technology
Semester : 6	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:4, T:0, P:0)	Periods per semester: 60

Course Objectives:

- To impart knowledge on the fundamentals of interior designing and relation with ergonomics.
- To familiarize the machines used for mass production of furniture.
- To familiarize the preparation of furniture for finishing and sanding.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Knowledge in Furniture joints		Workshop practice	3
Designing insight		Introduction to CAD Lab	3

Course Outcomes

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Interpret the basics in design and its relation with ergonomics	14	Understanding
CO2	Develop designing of residential spaces and public spaces	16	Applying
CO3	Explain furniture technology and machines used for mass production of furniture	16	Understanding
CO4	Illustrate the preparation of furniture for finishing.	12	Applying
	Series Test	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2				3		
CO2			3		2		
CO3				2			3
CO4			3		2		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Interpret the basics in design and its relation with ergonomics		
M1.01	Recall importance of Interior design and decoration	4	Remembering
M1.02	Relate ergonomics with interior design	4	Remembering
M1.03	Explain construction materials and its application	4	Understanding
M1.04	Describe Planning a project	2	Understanding
Content: Development of interior design- concepts - historic review. Principles of design - Balance, rhythm, emphasis, harmony, proportion - meaning and application of design concepts in the interior and exterior houses and other commercial buildings. Development of Ergonomics - Objectives - Approach to human factors - system- man machine system - system reliability Display - Type of Display, selection, compatibility Process of seeing - visual - visual compatibilities - Graphic representation of data -symbols - Quantitative reading - Qualitative reading - Quantitative reading - visual Display -Use of Anthropometric data. Building materials and finishes - Types and uses of stone, brick, timber, cement, mortar, concrete, plastics, glass, wood based materials, metals - ferrous and nonferrous, wall, floor and ceiling finishes			
CO2	Develop Designing of residential spaces and public spaces		
M2.01	Develop designing of Private space	4	Applying
M2.02	Develop designing of public spaces	4	Applying
M2.03	Explain Color Wheel and color theory	4	Understanding
M2.04	Illustrate wall treatment, window treatment and floor treatment	4	Applying
	Series Test-I	1	

Contents:

Planning and designing of interior spaces - planning considerations and drawings preparation of dining room, bed room, kitchen, bath room, library study room, storage area
 Planning and designing of interior spaces - design sinages and symbols. Free hand drawings

Concept of colour - significance of colour in the interiors and exteriors - Dimensions of colour - Colour wheel - Colour theory. Application of colour harmonies in the interiors and exteriors - Effects of light on colour, Illusion of colour, psychology of colour - General office multiple workstation - private offices - reception - hospitality spaces restaurants-hotel public spaces - private offices reception hospitality spaces - restaurants - hotel - public spaces - seminar hall - theatre - Auditorium - wall finishes floor and floor finishes-Wall treatment-Window Treatment and Floor Finishes.

Importance of lighting - Artificial lighting - light sources, Types and uses of light, specific factors in lighting - measurements of lighting and economy in lighting.

CO3	Explain furniture technology and machines used for mass production of furniture		
M3.01	Outline machines used for mass Production of Furniture	3	Remembering
M3.02	Identify portable machines	3	Remembering
M3.03	Develop Jigs and Fixtures used in furniture industry	3	Remembering
M3.04	Outline Estimating and costing	3	Applying
M3.05	Explain CNC Fundamentals	4	Remembering

Contents:

General consideration of furniture technology. History & development of furniture - Raw materials - preparation an characteristics - hand tools used in furniture making - Machines used for mass production of furniture (Brief ideas only) Circular saw, jig saw narrow band saw, thickness planner, jointer, spindle moulder, router, tenoning machine chain and chisel mortise, Doweling machine - Portable machines - portable drilling machine, portable jig saw, portable circular saw and sanders Saw doctoring - teeth geometry - tensioning, joining methods of saw - saw setting Furniture designing & production Fundamentals of design - design tools, planning a project, classification of joints - lengthening - widening, framing and box joints Jigs and fixtures Furniture hardware - hinges, screws, nails, locks, and accessories Preparation of bill of materials - estimating and costing of furniture-preparation of drawing of furniture

CO4	Illustrate the Preparation of Furniture For Finishing.		
M4.01	Outline Preparation of materials for finishing	6	Understanding
M4.02	Develop finishing of Furniture	6	Applying
	Series Test-II	1	

Contents:

Furniture finishing - preparation for finishing. Applications Sanding - coated abrasives - Stationery and portable sander. - Floor type belt sander- Disc sander - spindle sander, hand stroke belt sander. Portable sanders - finishing sanders, portable belt sanders, rotary action sanders important steps to be followed in Finishing- Staining and filling-sap staining - ingredients in stains-purpose of staining kinds of stain- water, oil and NGR stains- applying oil stains advantages and disadvantages (of above stains) and their ingredients. Wash coat - purpose - applying wash coat. Filling - kinds of fillers - paste and liquid fillers. Applying paste and liquid fillers. Sealing - applying a sealer Protective coating - oil finishers - linseed oil finisher - applying oil finish. Shellac - uses - types of Shellac- French polishing - Cellulose lacquers - ingredients. Advantages And disadvantages. Kinds of lacquers. - Lacquer thinners - lacquer finishing schedule - staining and filling before applying lacquer - sealers for lacquer - Applying a spray lacquer finish spraying a project- Applying a clear brushing lacquer finish. An introduction to plastic lacquers, polyurethane, polyester and epoxy finisher. Rubbing, polishing and cleaning - purpose

Text / Reference:

T/R	Book Title/Author
T1	Joseph De Chiala, Julius Pauero. Time Saver standards for interior Design and -, Space planning Martin Zelink, McGraw Hill, New York
T2	John A Walton, Wood works in theory and practice -, Australian publishing co.
T3	Robert Schalff, Complete Book of Wood finishing
R4	P.C Sharma ,Production Technology
R5	Mickomick, Human factors & Engineering

Online resources:

Sl. No	Website Link
1	https://wp.nyu.edu/dispatch/2019/03/26/how-technology-is-influencing-the-world-of-interior-design/
2	https://www.architecturaldigest.com/story/future-interior-design-profession
3	https://www.pinterest.com/dejoiedesign/science-and-technology-in-interior-design/
4	https://gatherit.co/how-tech-is-changing-the-business-of-interior-design/
5	https://www.jdinstitute.edu.in/technology-in-interior-design/
6	https://tdsblog.com/5-ways-technology-impacted-interior-design/
7	https://www.ied.edu/florence/design-school/undergraduate-courses/interior-and-furniture-design/DBH2976E